

The background features a large, solid black hexagon centered on the page. Surrounding this hexagon are various triangular shapes in vibrant colors: magenta, blue, red, yellow, green, cyan, and white. These shapes are arranged in a way that they appear to be fragments of a larger, colorful geometric pattern.

Colour Modes

KARL ABATE – EYESPARK.NET

This presentation is about the following colour modes:

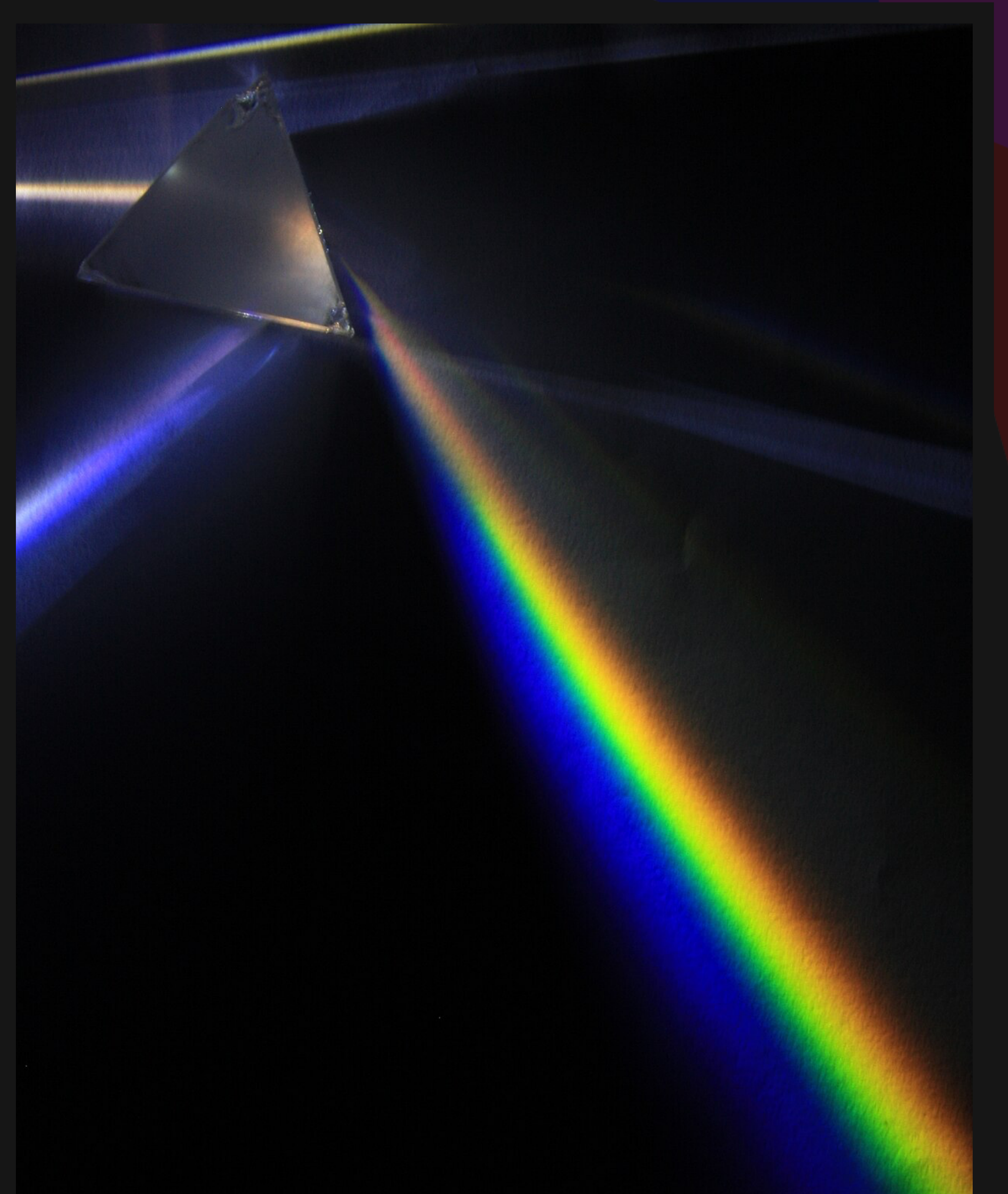
- CMYK
- RGB
- Hexadecimal
- HSB / HSV
- Escala de Grises
- Indexado
- Duotono
- CIE LAB

How Colours Work

Before learning about colour modes, we need to understand how colours themselves work.

The range of the electromagnetic spectrum that we are able to see with our eyes only has specific colours. It goes from red light (750 nm) to violet light (400 nm), with others in between.

When specific wavelenghts combine they can create different colours, including ones that aren't part of that range, including purple, pink, and white.

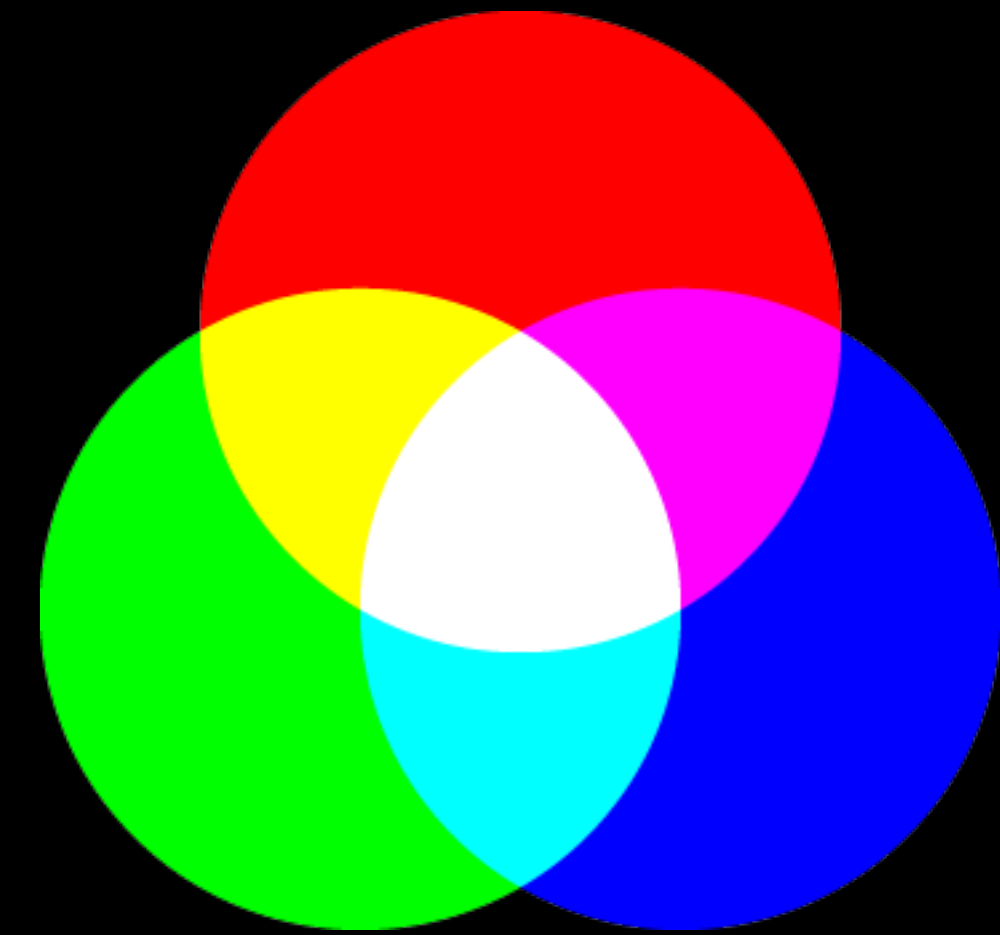


How Colours Mix

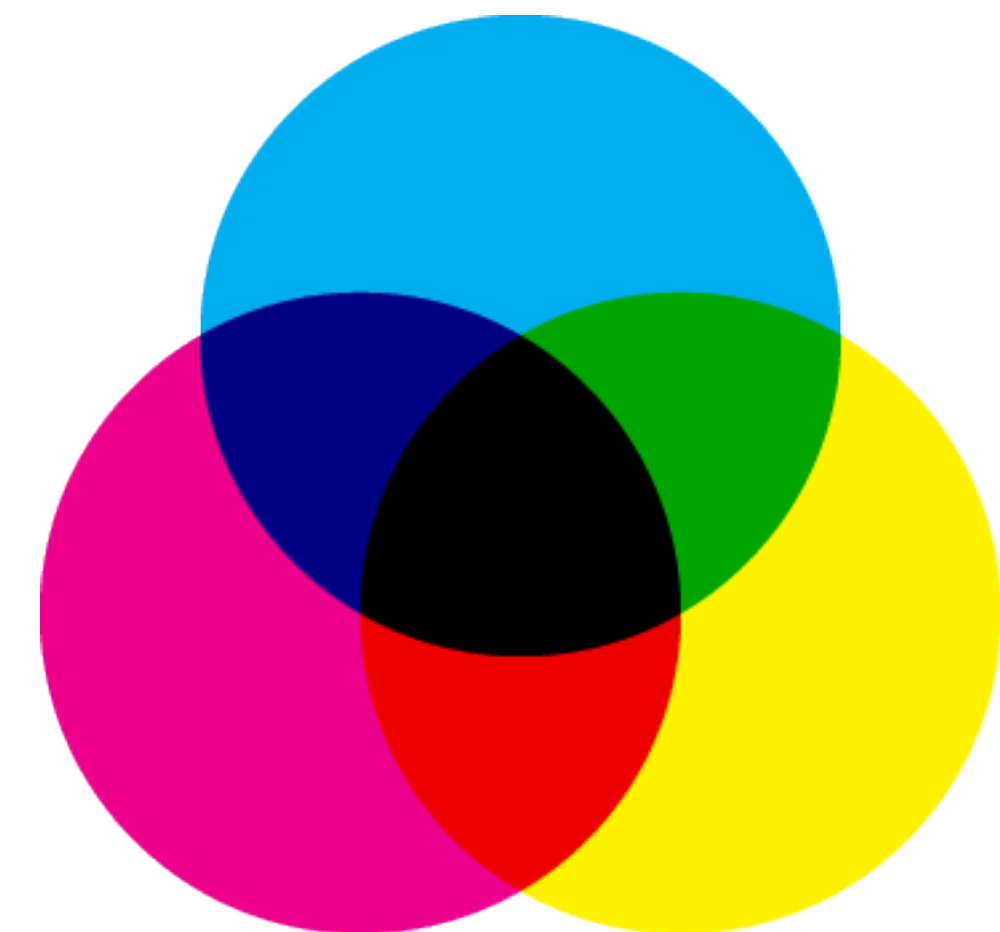
There are two ways to mix colours: subtractive and additive.

Light works additively, meaning that as different colours of light are added they turn brighter. All visible colours can be simulated using only three wavelengths: red, green, and blue.

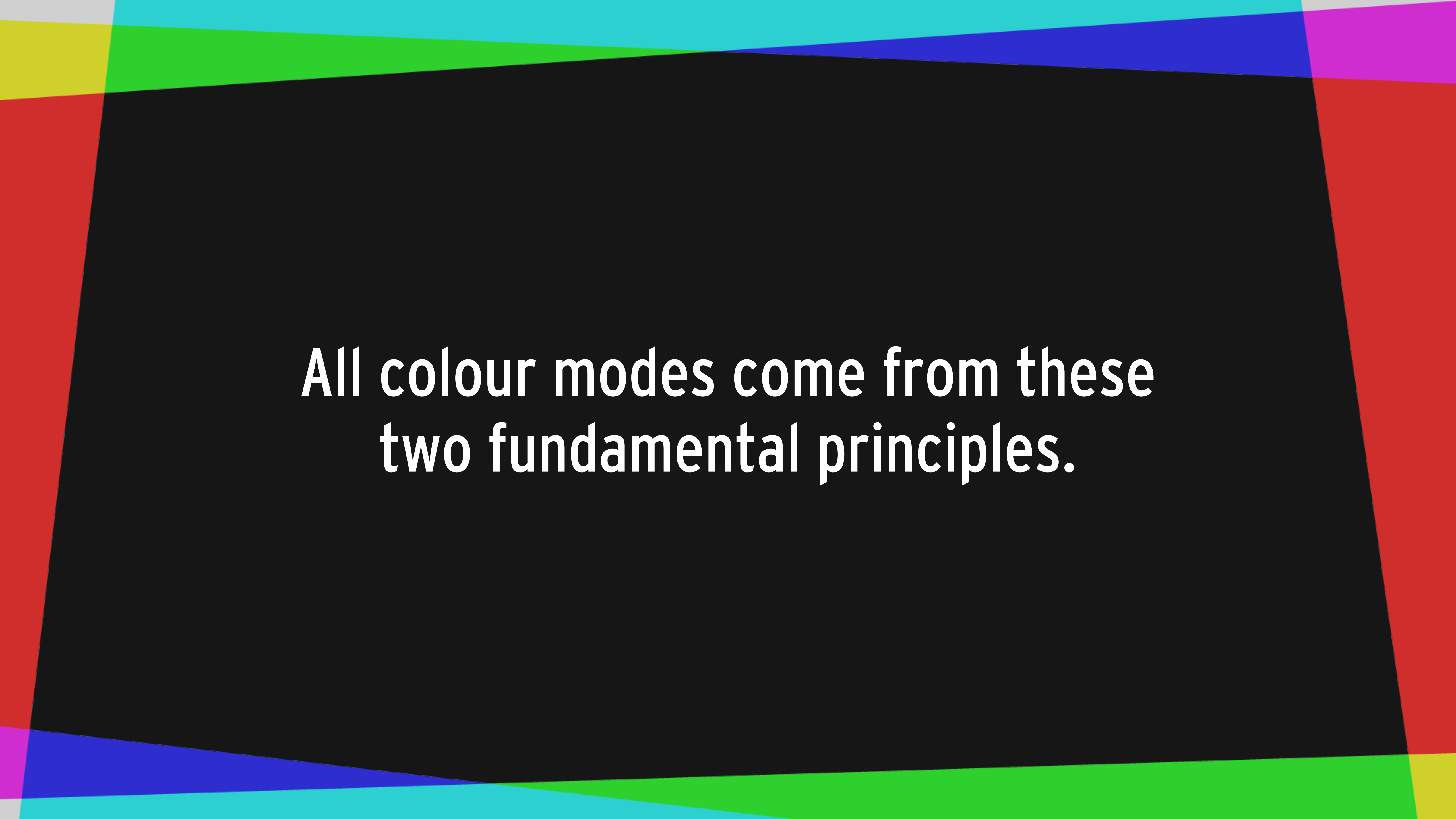
Pigments work subtractively. The primary pigments all absorb one of the three primary colours of light, resulting in cyan, magenta, and yellow, that all combine to make black.



Additive Mixing



Subtractive Mixing

The background features a large, solid black rectangle that occupies most of the frame. This rectangle is surrounded by various colorful geometric shapes, including triangles and polygons in shades of red, yellow, green, blue, and cyan, which are positioned at the corners and edges of the image.

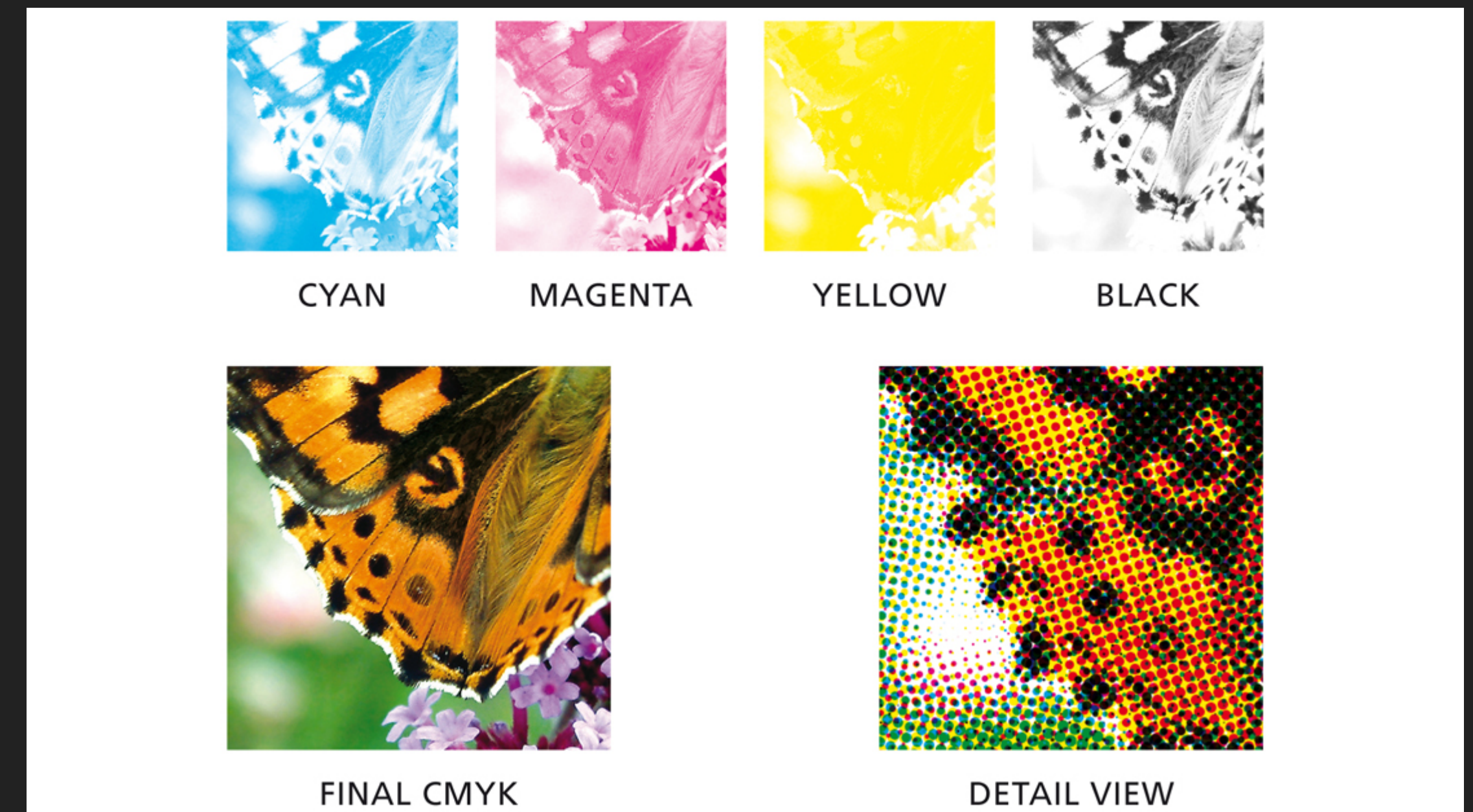
**All colour modes come from these
two fundamental principles.**

CMYK Mode

The CMYK colour mode is the main subtractive mode, that is used with everything involving pigments.

This uses the opposites of the three primary colours of light mentioned previously: cyan, magenta, and yellow, and also uses an additional black ink as a key. The base (0, 0, 0, 0) is white.

This mode is particularly common and noticeable with printers, which use a range of 0% - 100% to represent how much of each colour is in one spot.

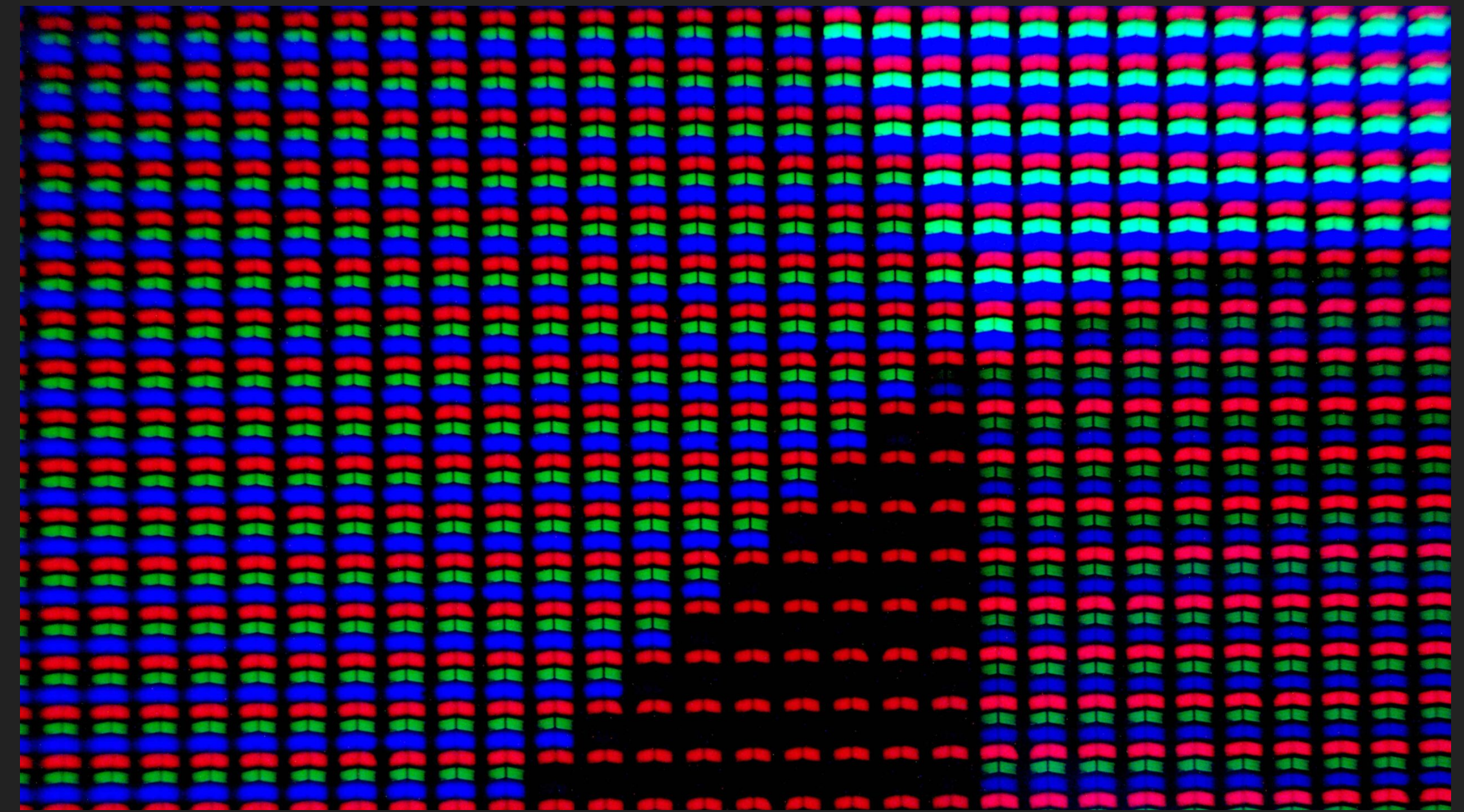


RGB Mode

The RGB colour mode is the main additive mode, that is used with everything that involves light itself.

This uses the colours at the extremes of the visible spectrum, red and blue, and green in the middle. The base (0, 0, 0) is black.

Since it works with light directly it's used in artificial lights and screens. Pixels on a screen emit all three colours, and their brightness can be adjusted to create different colours!



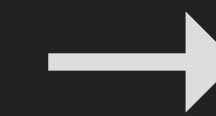
Hexadecimal Notation

Most screens now process 8 bits of information, which means that every sub pixel has a range of 256 levels of brightness (2^8), from 0 - 255.

These numbers can be represented in the standard base-10 format (decimal), but a more efficient way is using hexadecimal (base-16), which has the digits from 0 - 9 and adds the letters A - F to represent the numbers from 10 - 15.

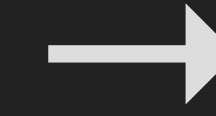
This system can represent 256 numbers with two digits, and every colour with six.

22, 22, 22



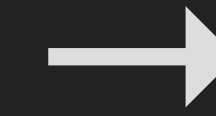
#161616

191, 47, 240



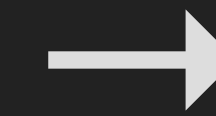
#bf2ff0

92, 164, 180



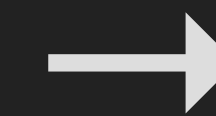
#5ca4b4

102, 204, 153



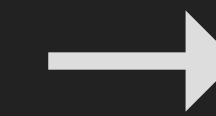
#66cc99

236, 172, 36



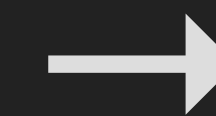
#ecac24

255, 0, 102



#ff0066

208, 222, 202



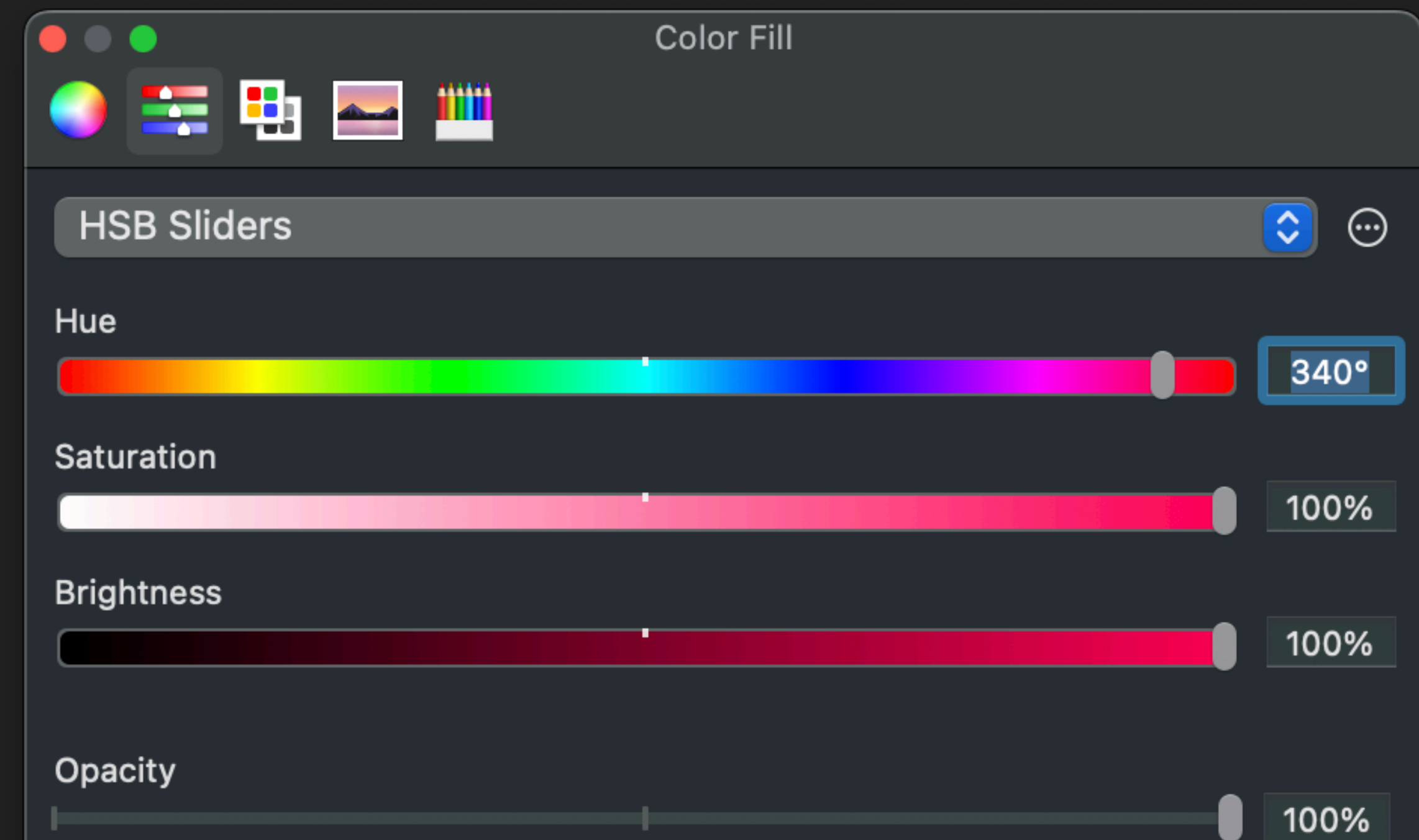
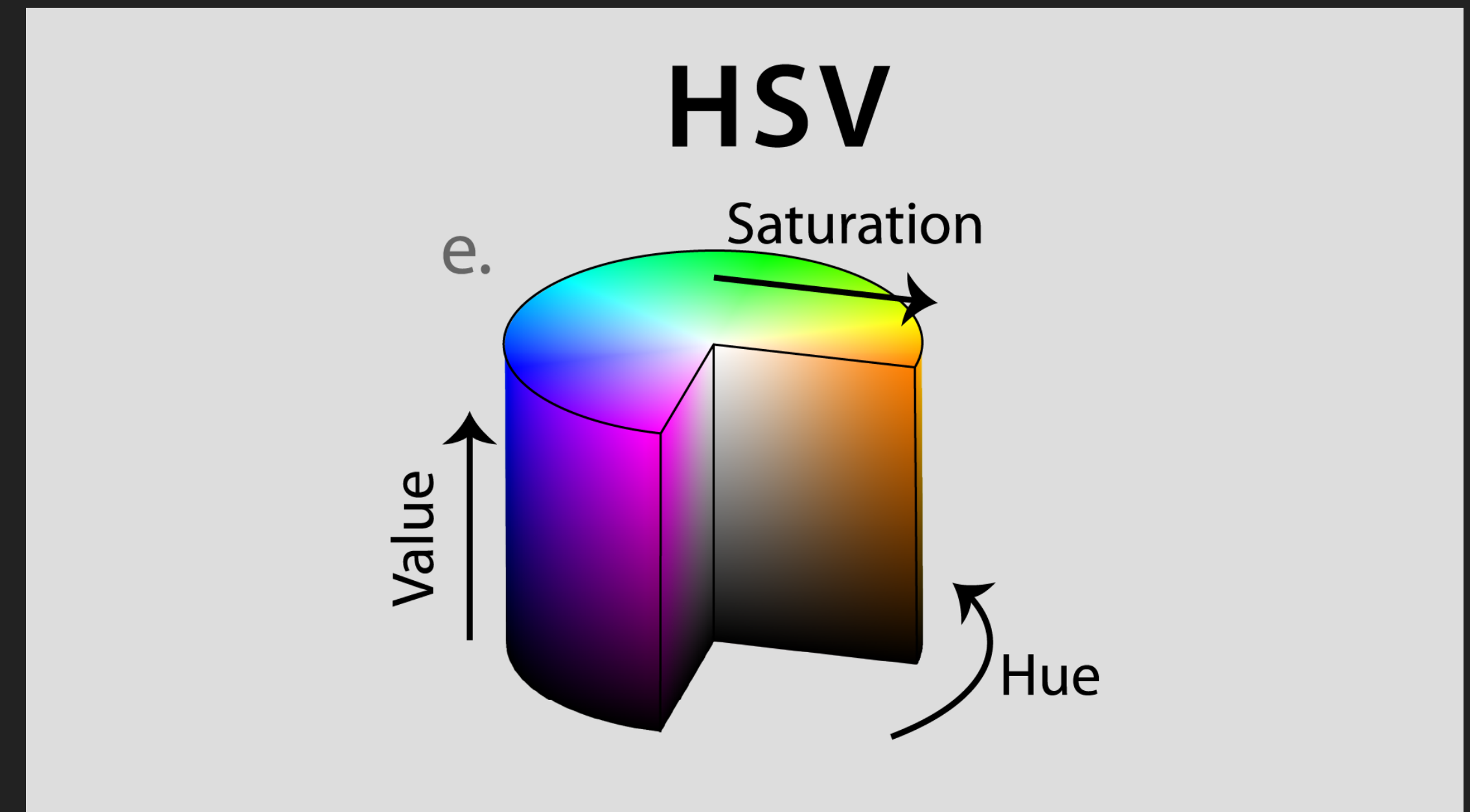
#d0deca

HSB/HSV Mode

The HSB or HSV colour mode is derived from the RGB mode, but formatted in a way that makes it more intuitive and comprehensible for us humans.

This mode contains three separate values. Hue, which contains all colours represented in 360°; Saturation, which represents how potent a colour is from 0% - 100%; and Brightness/Value, which represents how light or dark a colour is, also on a scale from 0% - 100%.

Not to be confused with HSL.



The following “modes” are
composed of sections from the
previous modes.

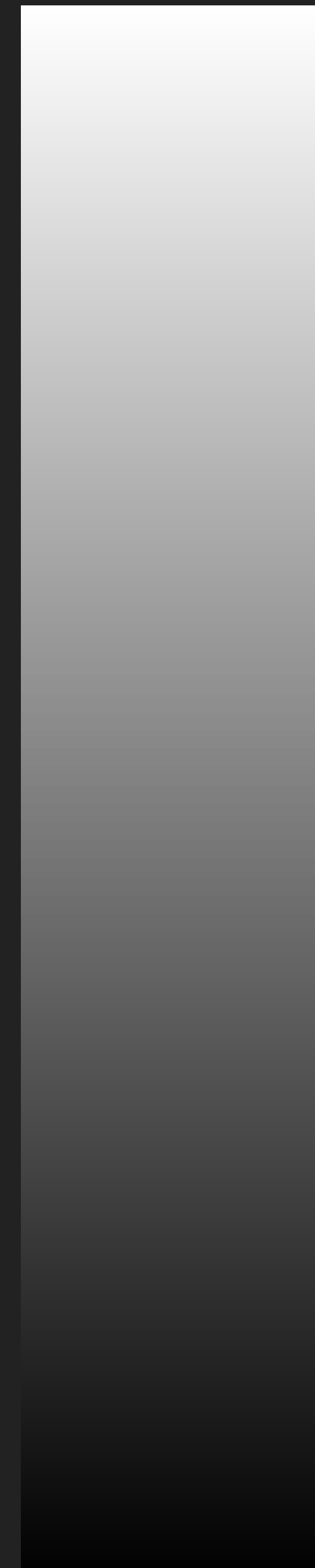
Grayscale

The grayscale is the section of the HSB and RGB colour modes that only contains the pure shades of grey.

In RGB that means that the red, green, and blue colour channels are the same. In HSB that means the saturation is at 0%.

There are different amounts of grey depending on the mode, 256 in RGB, and 101 in HSB, which is more common.

It can be used to process and create black and white images.



← 100% / 255, 255, 255 / #ffffff

← 90% / 230, 230, 230 / #e6e6e6

← 75% / 191, 191, 191 / #bfbfbf

← 60% / 153, 153, 153 / #999999

← 50% / 128, 128, 128 / #808080

← 40% / 102, 102, 102 / #666666

← 25% / 064, 064, 064 / #404040

← 10% / 026, 026, 026 / #1a1a1a

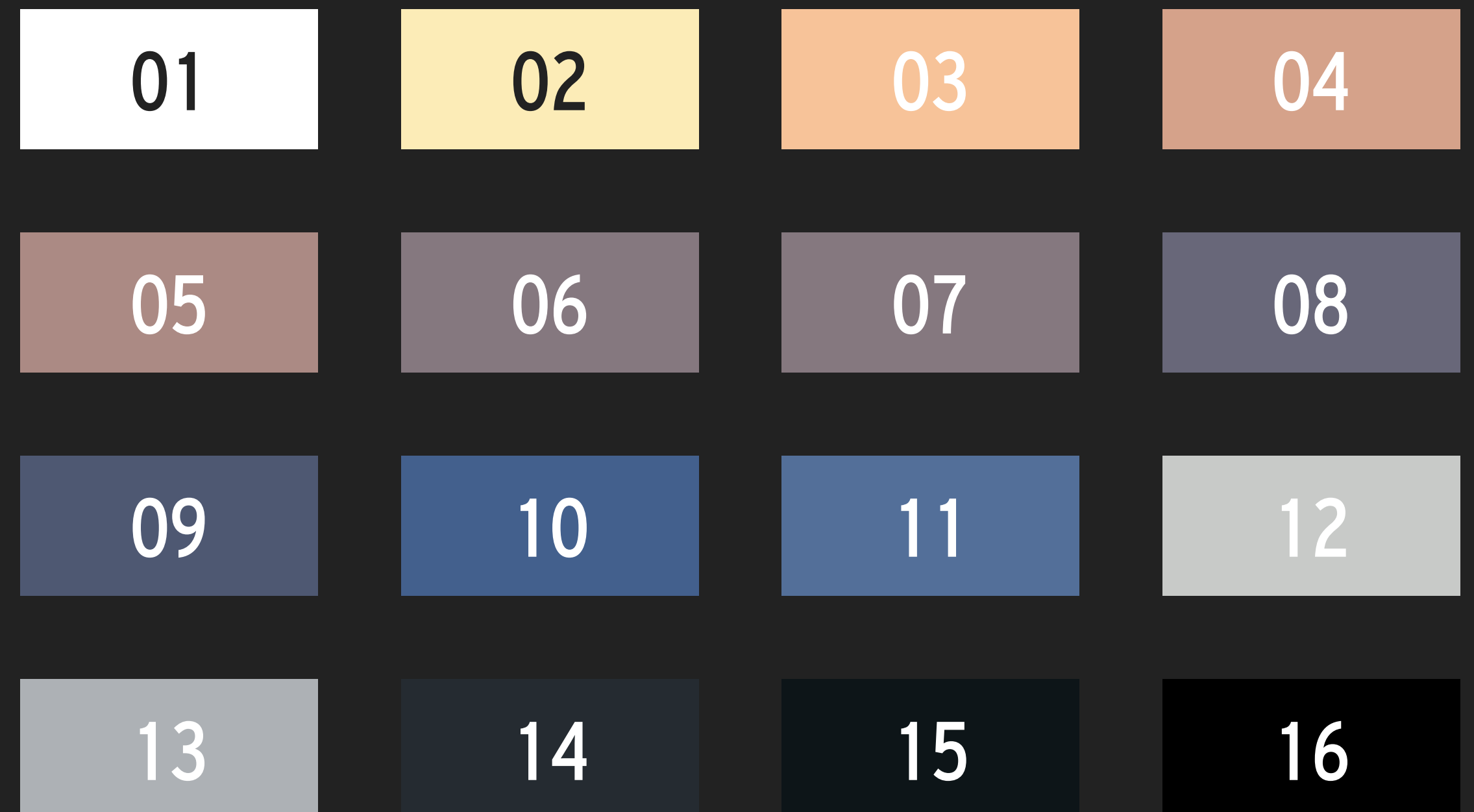
← 0% / 000, 000, 000 / #000000

Indexed Colours

Colours can be indexed to make them easier to process.

Instead of saving each specific colour of each specific pixel in each specific spot, file formats can assign specific colours a simple number, and use those in place of the full codes.

The quantity of colours and what numbers they belong to depend on which ones and how many are in the image, and that is heavily dependent on the desired quality of the final processed image.

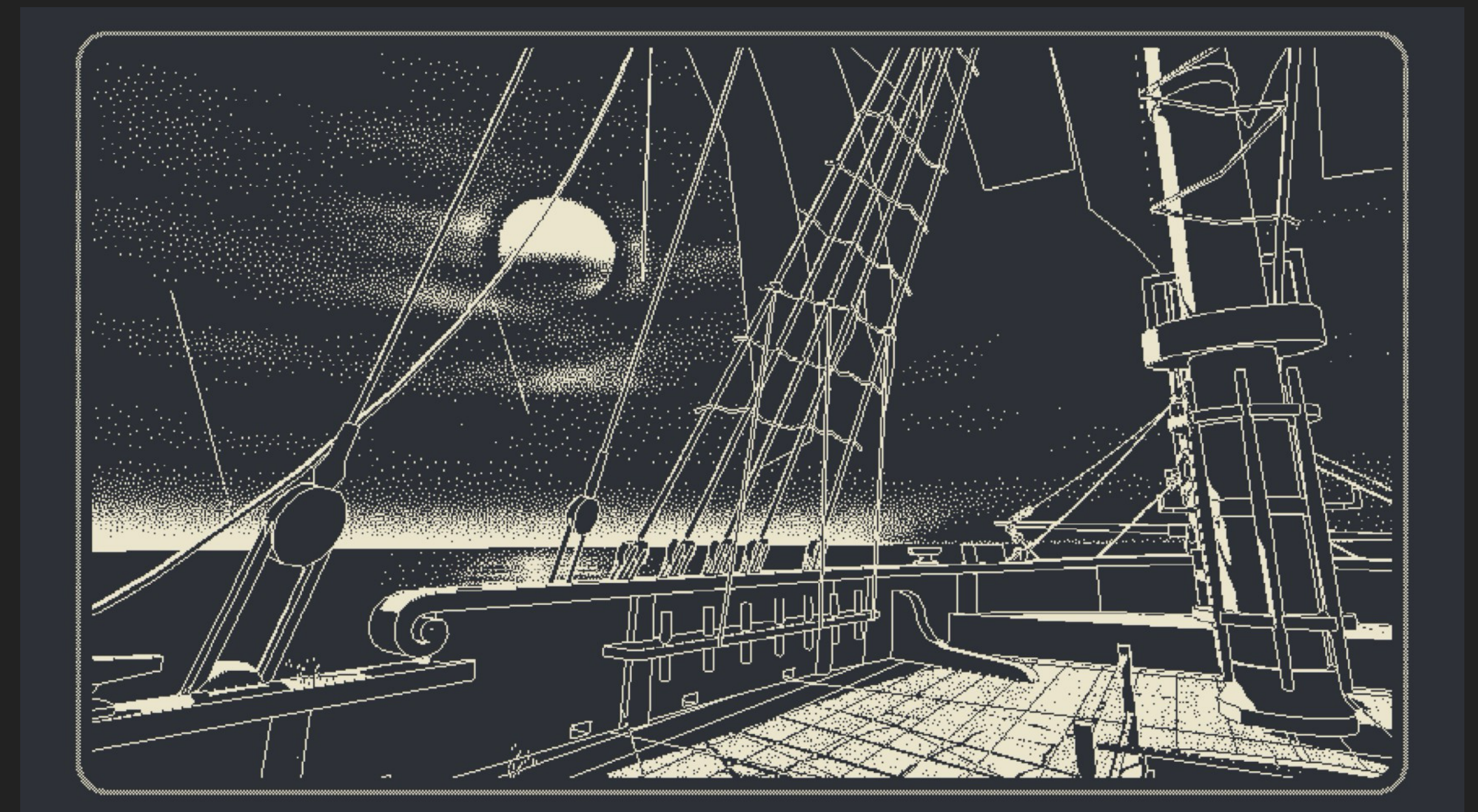
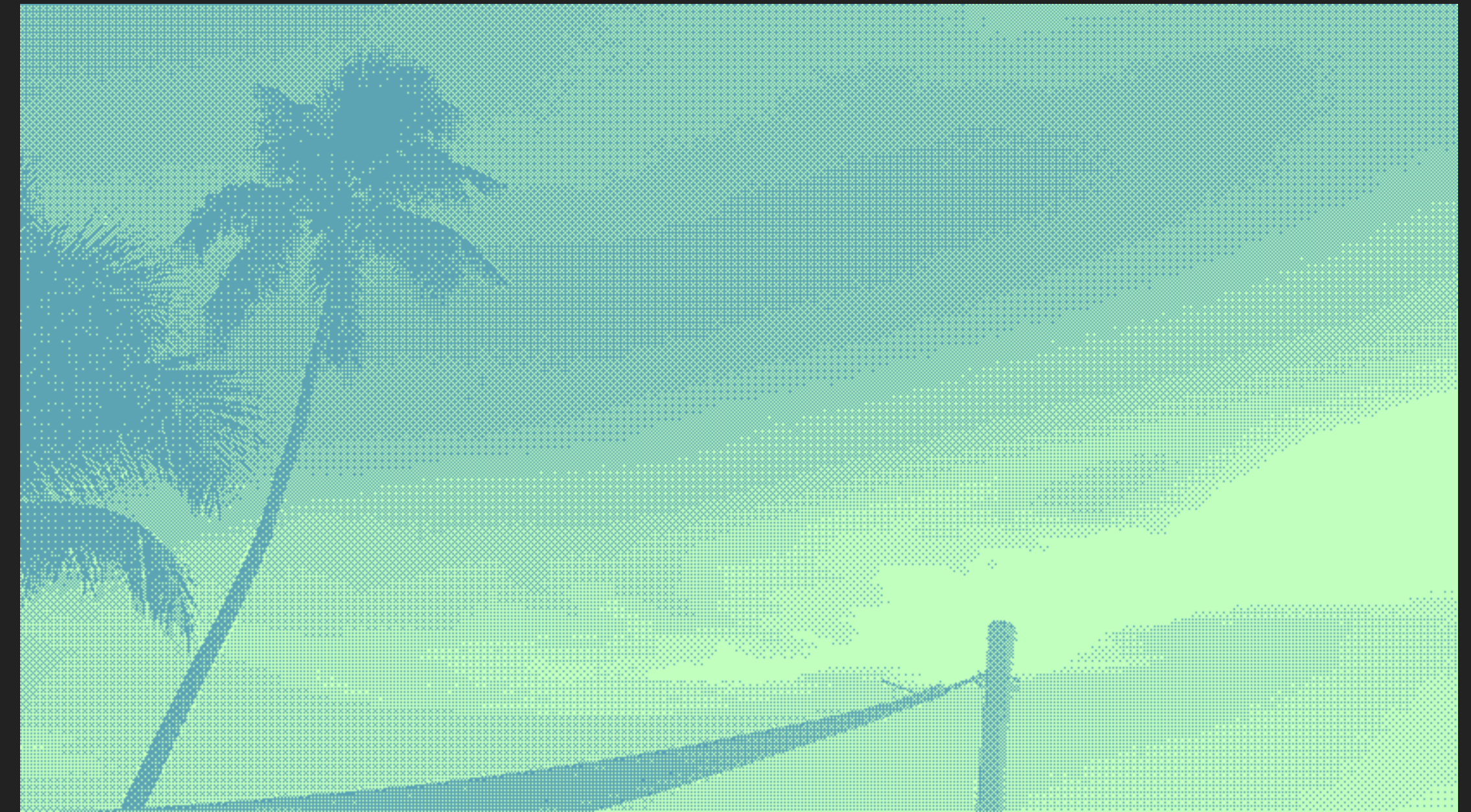


Duotone Images

In the digital context, the duotone style is a modification to an image that makes it so the entire thing is composed of pixels of only two colours.

This is different from the monotone style, which looks similar but is composed differently. Monochromatic images have only one hue but have all possible shades of it, bright and dark.

True duotone images only use two very specific colours, and nothing else. Every pixel is either one colour, or the other.

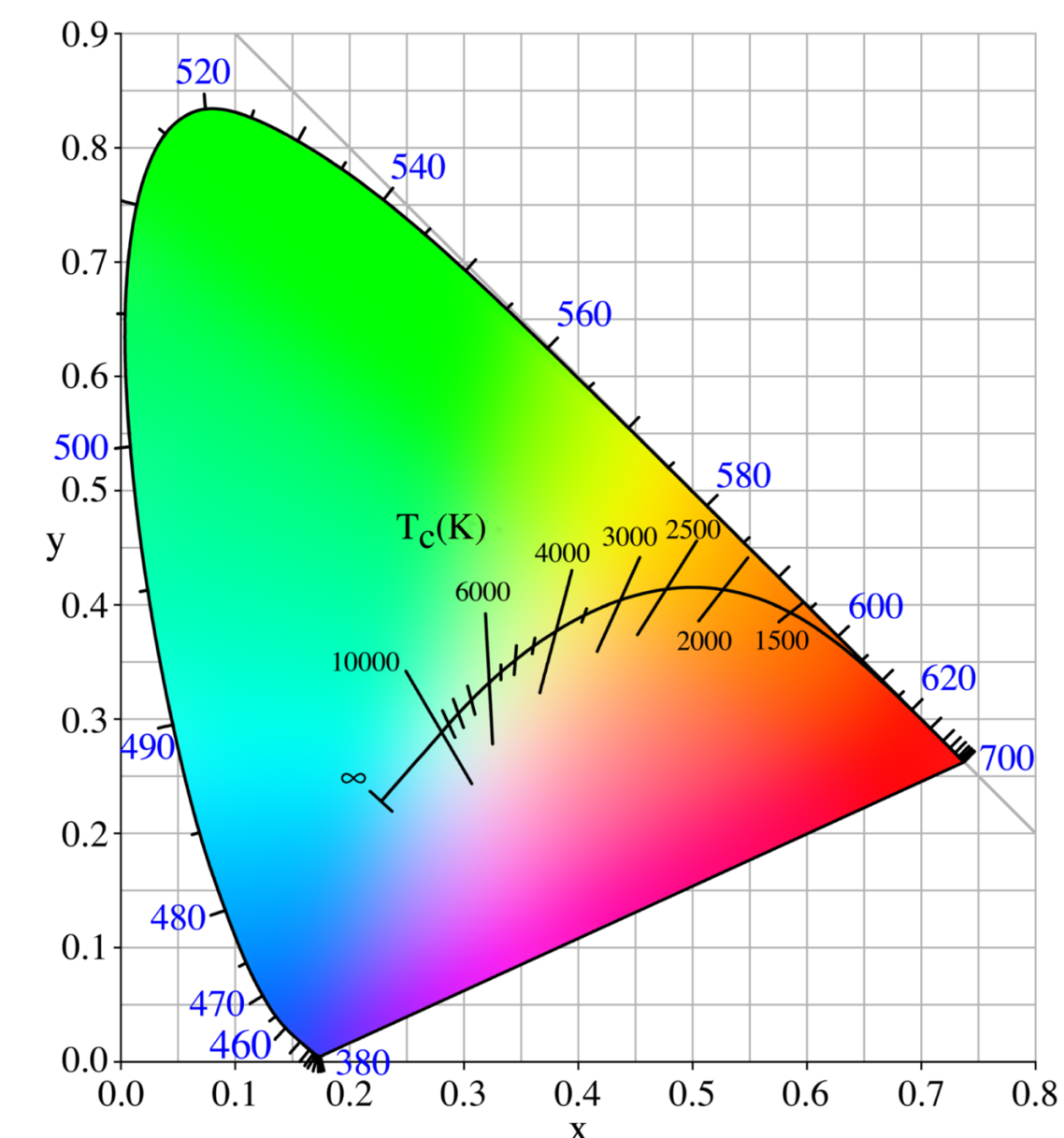
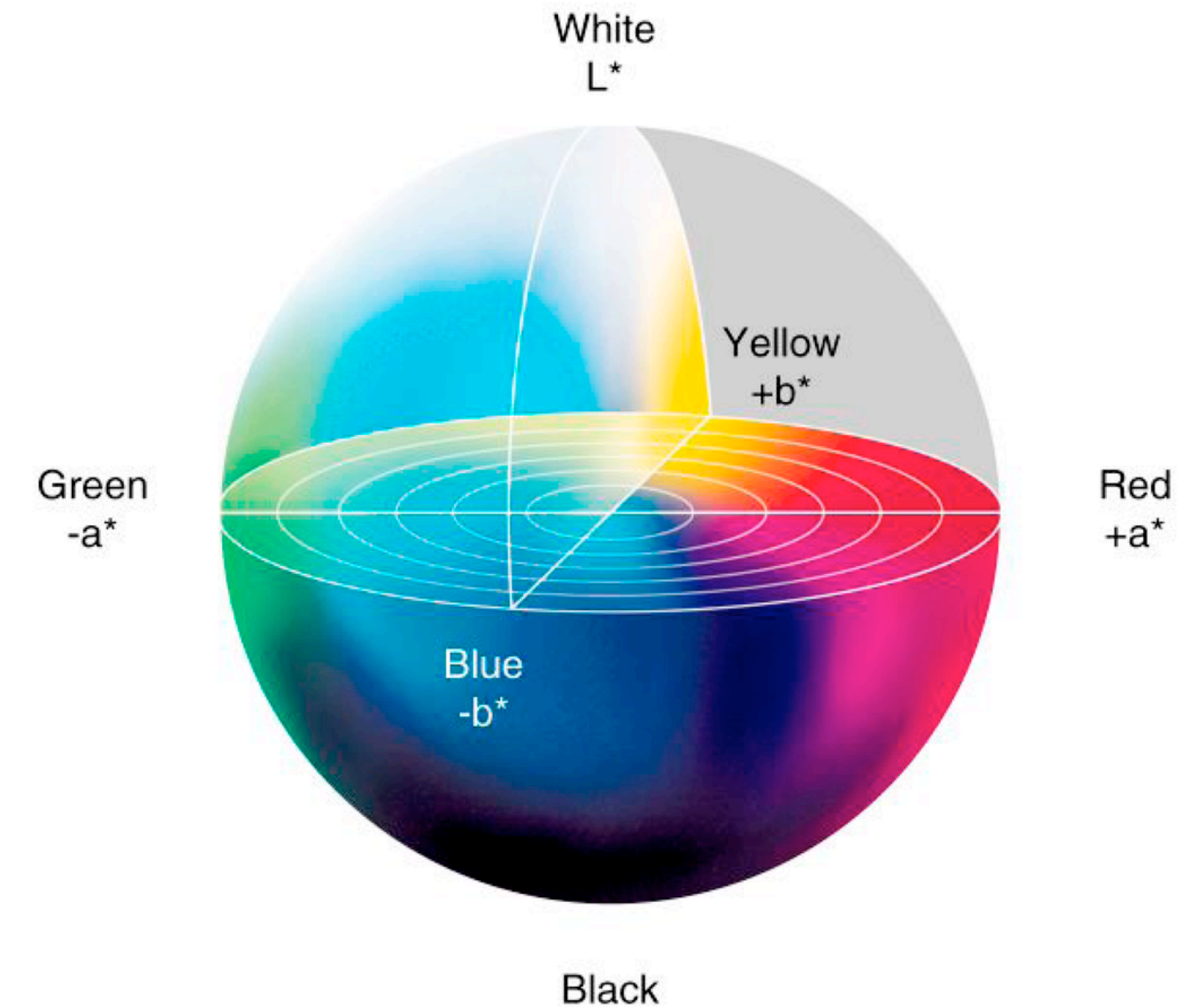


CIE LAB Space

The CIE LAB colour space was created by the International Commission on Illumination to represent colours in a way more similar to how we, humans, process colours in our brains.

The simplest way to visualize it is through a sphere. The hues are in a great circle, lightness goes from pole to pole, and saturation goes from the core.

This space doesn't have any real practical use, but is used at the technical level to change colours between modes.



The background features a large, solid black hexagon centered on the page. Surrounding this hexagon are several colorful triangles in shades of blue, green, yellow, red, and magenta, creating a vibrant, abstract geometric pattern.

Thank You!

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Bibliografy

- Lynch, David K, et al. "Color and Light in Nature". Cambridge University Press, 11 June 2001.
- minutephysics. "This Is Not a Rainbow." YouTube, 28 Oct. 2014, <https://www.youtube.com/watch?v=9udYi7exojk>.
- "Why Do Printers Use Cyan and Magenta instead of Blue and Red?" ETG Design, 2025, <https://etgdesign.com/cmyk/>.
- Scott, Tom. "Why Dark Video Is a Terrible Blocky Mess." YouTube, 30 Mar. 2020, <https://www.youtube.com/watch?v=h9j89L8eQQk>.
- Bear, Jacci Howard. "Understanding the HSV Color Model." Lifewire, 29 July 2020, <https://www.lifewire.com/what-is-hsv-in-design-1078068>.
- "Color Indexado." Academia-Lab.com, 2024, <https://academia-lab.com/enciclopedia/color-indexado/>.
- "About Color." Adobe.com, 2023, <https://helpx.adobe.com/illustrator/using/color.html>.

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